

Start coding or generate with AI.

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```
import numpy as np

# eg 1
ar=np.array([[12,3,6],
             [4,335,64]])
print(ar.size)
print(ar.ndim)

# eg 2
# eye-> gives identity matrix
np.eye(3)
```

```
6
2
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
ar=np.array([[12,3,6],
             [4,335,64]])

# 1
print(ar.sum()) #give sum of whole array

# 2
print(ar.sum(axis=0))#add vertically

# 3
print(ar.sum(axis=1))#add horizontally
```

```
424
[ 16 338  70]
[ 21 403]
```

```
#reshape
ar = np.array([10, 20, 30, 40, 50, 60])
print(ar.reshape(-1, 3))

ar2 = np.array([1, 2, 3, 4, 5, 6, 7, 8])
print(ar2.reshape(2, 2, 2))

arr = np.array([5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60])
k = arr.reshape(3, 4)
print(k)
```

```
[[10 20 30]
 [40 50 60]]
[[[1 2]
  [3 4]]

 [[5 6]
  [7 8]]]
[[ 5 10 15 20]
 [25 30 35 40]
 [45 50 55 60]]
```

```
ar=np.array( [[12,3,6,3,56,78],  
              [4,335,64,65,78,97]])  
demo=ar.ravel()
```

```
arr=np.array( [12,3,6,4,33,64,54,82,84,2,98,87])  
k= arr.reshape(3,4)  
arr[0]=100  
\
```

```
arr2=np.array( [12,3,6,4,33,64,54,82,84,2,98,87])  
k2= arr2.reshape(3,4).copy()
```

```
#Broadcasting  
d=np.array( [100,23,45,67,89,98,87,65,43,32,56,78])  
print (d*3)
```

```
n=[5,6,7]  
print(n*3)
```

```
#Masking  
mask= (d>80)  
print(mask)  
print(d[mask])
```

```
print(d[~mask])
```

```
a= (d>70)&(d<90)
print(d[a])
```

```
m=(d%2==0)|(d%5==0)|(d%7==0)
data= d[~m]
print(data)
```

```
[300  69 135 201 267 294 261 195 129  96 168 234]
[5, 6, 7, 5, 6, 7, 5, 6, 7]
[ True False False False  True  True  True False False False False]
[100  89  98  87]
[23 45 67 65 43 32 56 78]
[89 87 78]
[23 67 89 87 43]
```

```
marks=np.array([50,30,40,98,100,70,80,87,98])
condition = [
    marks>=90,
    marks>=70,
    marks>=50,
    marks>=33
]
```

```
choice = ['A', 'B', 'C', 'D']
```

```
print(np.select(condition, choice, default='Fail'))
```

```
['C' 'Fail' 'D' 'A' 'A' 'B' 'B' 'B' 'A']
```

```
import pandas as pd
```

```
#dataframe
d1 = {'Name': ['Aman', 'Rohit', 'Karan'], 'Marks': [85, 90, 75]}
df1 = pd.DataFrame(d1)
print(df1)
```

```
d2 = [[1, 'Apple'], [2, 'Banana'], [3, 'Mango']]
df2 = pd.DataFrame(d2, columns=['ID', 'Fruit'])
print(df2)
```

```
df3 = pd.DataFrame()
df3['Roll'] = [101, 102, 103]
df3['Pass'] = [True, False, True]
print(df3)
```

```
   Name  Marks
0  Aman     85
1  Rohit    90
2  Karan     75
   ID  Fruit
0   1  Apple
1   2  Banana
2   3  Mango
   Roll  Pass
0   101   True
1   102  False
2   103   True
```

```
#head
data = {'A': [1, 2, 3, 4, 5, 6, 7], 'B': [10, 20, 30, 40, 50, 60, 70]}
df = pd.DataFrame(data)
```

```
print(df.head())
```

```
print(df.head(2))
```

```
print(df.head(6))
```

```
   A  B
0  1 10
1  2 20
2  3 30
3  4 40
4  5 50
   A  B
0  1 10
1  2 20
   A  B
0  1 10
1  2 20
2  3 30
3  4 40
4  5 50
5  6 60
```

```
#sort
marks_data = {'Name': ['Aman', 'Rohit', 'Karan', 'Rahul', 'Jay'], 'Marks': [85, 90, 75, 60, 95]}
df_sort = pd.DataFrame(marks_data)

print(df_sort.sort_values(by='Marks'))

print(df_sort.sort_values(by='Marks', ascending=False))

print(df_sort.sort_values(by='Name'))
```

	Name	Marks
3	Rahul	60
2	Karan	75
0	Aman	85
1	Rohit	90
4	Jay	95

	Name	Marks
4	Jay	95
1	Rohit	90
0	Aman	85
2	Karan	75
3	Rahul	60

	Name	Marks
0	Aman	85
4	Jay	95
2	Karan	75
3	Rahul	60
1	Rohit	90

